ME 207: Fluid Dynamics

Professor:- Dilip Srinivas Sundaram, Uddipta Ghosh

Lab Experiment 7: Major and Minor Losses

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Group 14

Atharva Suryarao	23110051
Kedar Kanzariya	23110156
Khandvi Parth Piyushbhai	23110167
Jainil Maru	23110200
Paras Shirvale	23110232

Abstract

The experiment aims to analyze the losses in pipes and also comment on the importance of friction factors while analyzing the flow through pipes.

1 Task 7.1 (Essential Background)

Q1: What are Major and Minor head losses in a flow? Why is it important to quantify these losses?

Major losses occur mainly due to energy loss caused by friction between the fluid and the pipe walls. The more friction between the pipe and the fluid, the more significant losses will be, meaning the pressure drops significantly, signifying energy loss and vice versa.
A fluid passing through piping systems such as valves, bends, elbows, tees, inlets, expansions, and contractions incurs losses as these components interrupt the smooth flow of the fluid. The fluid, while going through sections like bends and elbows, leads to a change in the direction of the fluid, which leads to momentum change, resulting in energy loss. Similarly, fluids passing through contractions and expansions generate small vortices in the flow, decreasing the flow's energy. These types of losses are termed as minor losses.
It is important to quantify these losses because unless we know about the energy losses in flow due to major and minor losses, we will not be able to calculate the power of the pump required to make the fluid rise to particular heights; hence, it is essential to consider these factors in our calculations.

Q2: How does the major and minor losses depend on the flow rate?

The formula for calculating major head loss is:-

$$H = 4C_f \frac{Lu^2}{2gD}$$

- C

As head loss is proportional to the square of velocity, we can conclude that the major head loss is also proportional to the square of the flow rate.

$$H \propto u^2 Q^2$$

The formula for calculating minor head loss is:-

$$H = k\frac{u^2}{2g}$$

As head loss is proportional to the square of velocity, minor head loss is also proportional to the square of flow rate.

$$H \propto u^2 Q^2$$

Q3: What is Friction coefficient? How is Minor loss coefficient defined?

The friction coefficient is the resistance to the fluid flow due to friction between the pipe walls and fluid. It is defined only for major losses.
For Minor losses, the coefficient is defined as the energy lost to sudden contractions and expansions and changes in direction for flow. The coefficient (K) lies between (0-1), and its value must be calculated experimentally.

$$H = k\frac{u^2}{2g}$$

Q4: How are head losses quantified in Valves, contractions and expansions, pipe bends and elbows?

The head loss in valves, contractions and expansions, pipe bends and elbows are termed as minor losses. They are calculated using the following formula:-

$$h_f = K\frac{u^2}{2g}$$

Where:

- h_f = head loss (m)
- K = loss coefficient (depends on the type of fitting)
- V = velocity of fluid (m/s)
- g = gravitational acceleration (9.81 m/s²)

Q5:What does a Moody's chart do?

Moody's chart presents the darcy friction factor for pipe flow as a function of Reynolds number and $\frac{\epsilon}{D}$ over a wide range. It helps calculate the friction factor in pipe flow through a graphical representation obtained by conducting experiments. It is probably one of the most widely accepted and used charts in engineering. Although it is developed for circular pipes, it can also be used for noncircular pipes by replacing the diameter with the hydraulic diameter.
Where, $\frac{\epsilon}{D}$ is the relative roughness.

Q6: What is friction factor and what does it depend on?

In the context of major losses, the friction factor (f) is essentially the kinetic energy coefficient as it is a factor lying between 0 - 1; it tells us about the energy lost to friction, so we can say that it indeed is the kinetic energy coefficient.

2 Introduction

Figure 1: Image of the experimental setup

In fluid Dynamics, whenever a liquid flows through the pipe energy is lost and this loss is divided into two categories namely Major losses and Minor losses. Generally, loss due to the friction while water flowing in the pipe in considered as mojor loss. And losses occured due to the valves, bends, contractions, and the fitting are known as minor losses. Due to this we need to to have an effective flow system to reduce this high losses to reduce to energy consumption.

This experiment focuses on measuring both major and minor losses in a controlled pipeline system. Water is passed through pipes made of different materials, passing through various fittings, and the pressure drop at specific points is recorded using manometers. So, we are going to observe how flow rate affects the losses by performing the experiment with different flow rates.

Additionally, we will calculate loss coefficients for different minor loss components and determine their equivalent lengths, comparing them with standard reference values. The impact of pipe material on frictional losses will also be examined to assess how surface characteristics influence resistance to flow. So, we will evaluate uncertainties in measurements and analyze potential sources of error to ensure the reliability of our findings.

Through this experiment, we will gain practical insights into head losses in piping systems and their significance in real-world engineering applications.

Below attached is the schematic diagram of the apparatus.

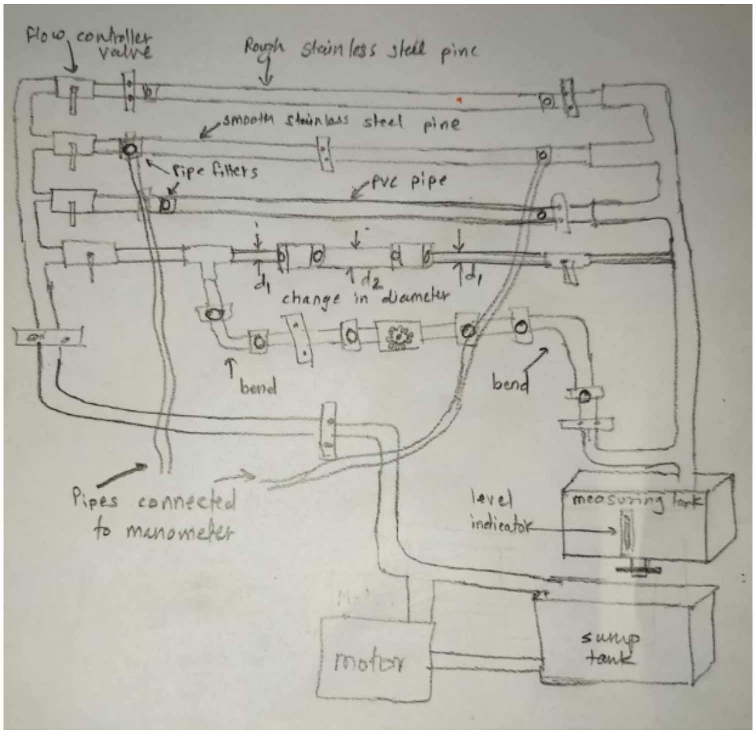

Figure 2: Schematic diagram of the apparatus

3 Objectives

(i) Measure Major losses in a straight tube for various flow rates.
(ii) Measure Minor losses in bends, valves, contractions etc., for various flow rates
(iii) Quantify the loss coefficient in various minor loss components.

4 Experimental Procedure

1. Before starting the experiment, it is important to ensure that the sump tank is filled with sufficient water so that the amount of air bubbles in the flow can be minimized.

2. The flow rate could be controlled by using the valve present toward the end of the main pipe, which is right above the measuring tank, as shown in the image. Initially, the flow rate is set to be maximum.

3. After setting the flow rate to its maximum value, the time for the water level to rise by 2 cm in the measuring tank is measured, and from that, using the known dimensions of the measuring tank, the flow rate is calculated. This was done using a valve that controls the flow of water across the sump tank and the measuring tank.

4. Before taking the reading in manometers, the bleeding has to be done, after which the air is pumped into it, and the height difference between both columns is noted. This is repeated to take three readings from each manometer.

5. Then, two manometers are disconnected from the rough steel pipe and PVC pipe and are connected to the fourth pipe, where the flow undergoes sudden contraction and sudden expansion. Measurements are done in the same way as those for PVC pipe and steel pipes.

6. Manometers are then disconnected from the fourth pipe and connected across the regions in the fifth pipe, where the flow undergoes an elbow bend and a long bend. Values of height differences are again noted using a procedure similar to the one mentioned in the fourth step.

7. Now, the flow rate is changed and measured again using the method indicated in the third point. The entire process, starting from the fourth step, is repeated to measure the values of height difference for the second flow rate, which indicates losses.

8. Now, keeping the flow rate the same, one manometer is connected across the gate valve, and the height difference is measured as mentioned in the fourth step.

9. Finally, the flow rate is changed to maximum, and measurements are taken again across the gate valve as done in the above step.

5 Results and Error Analysis

We did this experiment for two flow rates. First, we'll find the volumetric flow rate for both readings. To find the volumetric flow rate, we'll find the time taken by the fluid (water in this case) to reach a certain height in the container. We'll take the height of two centimeters in both readings. Then we'll divide the volume of the container by that time to find the volumetric flow rate.

Calculation of the flow rate

- **For the first flow rate**

We take the height of 2 centimeters in a container to measure the volumetric flow rate. The volume of the container(or measuring tank) is $30\,\text{cm} \times 50\,\text{cm} \times 2\,\text{cm} = 3000\,\text{cm}^3$.

Reading number	**Time taken (sec)**	**Total volume (cm^3)**	**Volume flow rate (cm^3/s)**
1	12.69	3000	236.406
2	13.84	3000	216.763
3	14.24	3000	210.673
Time Average		**13.59±0.6**	
Flow rate Average			**221.281±10.083**

- **For the second flow rate**

We take the height of 2 centimeters in a container to measure the volumetric flow rate. The volume of the container(or measuring tank) is $30\,\text{cm} \times 50\,\text{cm} \times 2\,\text{cm} = 3000\,\text{cm}^3$.

Reading number	**Time taken (sec)**	**Total volume (cm^3)**	**Volume flow rate (cm^3/s)**
1	12.69	3000	236.406
2	13.84	3000	216.763
3	14.24	3000	210.673
Time Average		**13.59±0.6**	
Flow rate Average			**221.281±10.083**

5.1 Major Losses

Major losses are the losses due to friction. First, we'll find the velocity of the fluid from the value of the volumetric flow rate. Then using the Darcy-Weisbach equation, the value of the coefficient of friction (f) can be calculated from the formula given below:

$$h_f = \frac{fLv^2}{2Dg} \tag{1}$$

Where:

- h_f: Friction head loss/height difference in the manometer (m)
- D: Diameter of the pipe (m)
- g: Acceleration due to gravity (9.81 m/s^2)
- l: Length of the pipe (m)
- f: Friction factor ($f = 4C_f$, where C_f = Coefficient of friction)
- v: Mean velocity of the fluid (m/s)

5.1.1 Smooth Stainless steel pipe:-

The internal diameter of this pipe (D) is 2.7 cm, and the area of the pipe $\left(\frac{\pi D^2}{4}\right)$ is 5.728 cm^2. We'll find the friction factor using the Manometer readings. The length of the pipe (l) is 1m.

$$\text{Area of the pipe} = \frac{\pi D^2}{4}$$

$$= 5.728\ \text{cm}^2$$

- **For the first flow rate:**

- **Experimental:**

Reading Number	Left Tube (cm)	Right Tube (cm)	Height Difference (m)
1	56.2	59.7	0.035
2	88.9	91.9	0.030
3	89.7	92.4	0.027
Average			0.0306

Table 1: Manometer Readings and Height Differences

From the above table, we can determine the experimental friction factor by first determining the coefficient of friction using velocities from the volumetric flow rate.

Sr No.	Volume Flow Rate (cm^3/s)	Velocity (cm/s)	Velocity (m/s)	Height Difference (m)	Coefficient of Friction (C_f)	Friction Factor (f)
1	396.301	69.188	0.69188	0.035	0.009	0.038
2	446.428	77.9398	0.77939	0.030	0.006	0.026
3	417.827	72.9465	0.72946	0.027	0.006	0.026
Average	420.185	73.3583	0.73358	0.0306	0.007	0.030

Table 2: Experimental Friction Factor (First Flow rate)

- **Theoretical:-**

Now, we'll determine the theoretical friction factor using Moody's diagram. We'll take the relative roughness of this pipe 0.00 for this case. And the kinematic viscosity of the fluid is 1.01 x 10-6 m2/s. Using the formula given below, we'll find the values of Reynold's number for the different velocities that we obtained from different flow rates.

$$\text{Re} = \frac{VD}{v}$$

Where:

- V = Flow velocity (m/s)
- D = Internal diameter of the pipe (m)
- ν = Kinematic viscosity of the fluid (m^2/s)

Sr No.	Velocity (m/s)	Reynolds Number	Theoretical Friction Factor
1	0.691	1.85×10^4	0.0260
2	0.779	2.08×10^4	0.0255
3	0.729	1.95×10^4	0.0260
Average	0.733	1.96×10^4	0.0260

Table 3: Theoretical Friction Factor (First Flow rate)

- **For the second flow rate:**
- **Experimental:**

Reading Number	Left Tube	Right Tube	Height Difference (m)
1	90.4	92.0	0.016
2	92.3	94.2	0.019
3	94.0	95.0	0.010
Average			0.015

Table 4: Manometer Readings and Height Differences

From the above table, we can determine the friction factor by first determining the coefficient of friction using velocities from the volumetric flow rate.

Sr No.	Volume Flow Rate (cm^3/s)	Velocity (m/s)	Height Difference (m)	Coefficient of Friction (C_f)	Friction Factor (f)
1	236.41	0.413	0.016	0.0124	0.0498
2	216.76	0.378	0.019	0.0176	0.0703
3	210.67	0.368	0.010	0.0098	0.0392
Average	221.28	0.386	0.015	0.0133	0.0532

Table 5: Experimental Friction Factor (Second Flow rate)

- **Theoretical:-**

Now, we'll determine the theoretical friction factor using Moody's diagram. We'll take the relative roughness of this pipe 9.74 x 10-5 for this case. The kinematic viscosity of the fluid is 1.01 x 10-6 m2/s. Using the formula given below, we'll find the values of Reynold's number for the different velocities that we obtained from different flow rates.

$$\text{Re} = \frac{VD}{v}$$

Where:

- V = Flow velocity (m/s)
- D = Internal diameter of the pipe (m)
- ν = Kinematic viscosity of the fluid (m^2/s)

Sr No.	Velocity (m/s)	Reynolds Number	Theoretical Friction Factor
1	0.412	1.10×10^4	0.0275
2	0.378	1.01×10^4	0.0275
3	0.367	9.83×10^3	0.0310
Average	0.386	1.03×10^4	0.0306

Table 6: Theoretical Friction Factor (Second Flow rate)

5.1.2 Roughed Stainless Steel Pipe

The internal diameter of this pipe (D) is 1.576 cm, and the area of the pipe $\left(\frac{\pi D^2}{4}\right)$ is 1.951 cm^2. We'll find the friction factor using the Manometer readings. The length of the pipe (l) is 1 m.

$$\text{Area of the pipe} = \frac{\pi D^2}{4}$$
$$= 1.951 \text{ cm}^2$$

- **For the first flow rate:**
- **Experimental:**

Reading number	**Left tube (cm)**	**Right tube (cm)**	**Height difference (m)**
1	94.0	66.6	0.274
2	67.4	40.7	0.267
3	67.7	40.1	0.276
Average			0.272

Table 7: Manometer Readings and Height Differences

From the above table, we can determine the friction factor by first determining the coefficient of friction using velocities from the volumetric flow rate.

Sr No.	**Volume Flow Rate (cm^3/s)**	**Velocity (m/s)**	**Height Difference (m)**	**COF(C_f)**	**Friction Factor (f)**
1	396.301	2.030	0.274	0.005	0.020
2	446.428	2.287	0.267	0.003	0.015
3	417.827	2.141	0.276	0.004	0.018
Average	420.185	2.153	0.272	0.004	0.018

Table 8: Experimental friction factor (First Flow rate)

where, COF= Coefficient of Friction
Using Moody's diagram, There is no roughness value for this experimental friction factor and Reynolds number.

For the second flow rate:

Reading number	**Left tube (cm)**	**Right tube (cm)**	**Height difference (m)**
1	81.3	37.2	0.041
2	60	55.4	0.046
3	82.1	71.4	0.107
Average			0.064

From the above table, we can determine the friction factor by first determining the coefficient of friction using velocities from the volumetric flow rate.

Sr No.	Volume flow rate(cm^3/s)	Velocity(m/s)	Height difference(m)	COF(C_f)	Friction factor(f)
1	236.407	1.211	0.041	0.002	0.008
2	216.783	1.107	0.046	0.002	0.011
3	210.674	1.079	0.107	0.007	0.028
Average	221.281	1.134	0.064	0.003	0.015

where, COF= Coefficient of Friction
Using Moody's diagram, There is no roughness value for this experimental friction factor and Reynolds number.

5.1.3 PVC pipe:

The internal diameter of this pipe (D) is 1.54 cm, and the area of the pipe ($\pi D^2/4$) is 1.863 cm^2. We'll find the friction factor using the Manometer readings. The length of the pipe (l) is 1 m.

$$\text{Area of the pipe} = \left(\frac{\pi(1.54)^2}{4}\right) = 1.863\ \text{cm}^2$$

- **For the first flow rate**

Experimental:

Reading number	Left tube (cm)	Right tube (cm)	Height difference (m)
1	71.5	34.3	0.372
2	82.2	25.3	0.369
3	80.5	43.5	0.370
Average			0.370

From the above table, we can determine the friction factor by first determining the coefficient of friction using velocities from the volumetric flow rate.

Sr No.	Q (cm^3/s)	Velocity (m/s)	h (m)	COF(C_f)	Friction Factor (f)
1	396.301	2.12677	0.372	0.006	0.024
2	446.429	2.39577	0.369	0.004	0.019
3	417.827	2.24248	0.370	0.005	0.022
Average	420.186	2.25494	0.370	0.005	0.022

where, COF = coefficient of friction, Q = volume flow rate, h = Height Difference

Theoretical:

Now, we'll determine the theoretical friction factor using Moody's diagram. We will take the roughness of this pipe as 1.5E-06. We will take the relative roughness of this pipe as 9.74×10^{-4} for this case. The kinematic viscosity of the fluid is 1.01×10^{-6} m^2/s.

Using the formula given below, we'll find the values of Reynolds number for the different velocities that we obtained from different flow rates.

$$\text{Re} = \frac{VD}{\nu}$$

Where:

- V = Flow velocity (m/s)
- D = Internal diameter of the pipe (m)
- ν = Kinematic viscosity of the fluid (m^2/s) = 1.01E-06

Sr No.	**Velocity (m/s)**	**Reynolds number**	**Theoretical friction factor**
1	2.127	3.24×10^4	0.0235
2	2.396	3.65×10^4	0.0227
3	2.242	3.42×10^4	0.0230
Average	2.255	3.44×10^4	0.0231

For the Second Flow Rate:

Experimental:

Sr No.	**Left tube (cm)**	**Right tube (cm)**	**Height difference (m)**
1	79.4	66.0	0.134
2	74.6	60.9	0.137
3	84.2	70.5	0.137
Average	-	-	0.136

From the above table, we can determine the friction factor by first determining the coefficient of friction using velocities from the volumetric flow rate.

Sr No.	**Q (cm^3/s)**	**Velocity (m/s)**	**h (m)**	**COF(Cf)**	**Friction Factor**
1	236.469	1.269	0.134	0.0063	0.025
2	216.730	1.163	0.137	0.0076	0.031
3	210.674	1.130	0.137	0.0081	0.032
Average	221.2812692	1.187	0.136	0.0073	0.029

where, COF = coefficient of friction, Q = volume flow rate, h = Height Difference

Theoretical:

Now, we'll determine the theoretical friction factor using Moody's diagram. We will take the relative roughness of this pipe as 9.74×10^{-4} for this case. The kinematic viscosity of the fluid is 1.01×10^{-6} m^2/s.

Using the formula given below, we will find the values of Reynolds number for the different velocities that we obtained from different flow rates.

$$Re = \frac{VD}{v}$$

Where:

- V = Flow velocity (m/s)
- D = Internal diameter of the pipe (m)
- v = Kinematic viscosity of the fluid (m^2/s)

Sr No.	Velocity (m/s)	Reynolds number	Theoretical Factor
1	1.269	1.93×10^4	0.0263
2	1.163	1.77×10^4	0.0268
3	1.130	1.72×10^4	0.027
Average	1.187	1.81×10^4	0.0267

5.2 Minor Losses

Minor losses include losses due to the change in cross-section, valve, bend, sudden compression, sudden expansion, and the elbow. First, we'll find the velocity of the fluid from the value of the volumetric flow rate. The minor losses are calculated using the formula,

$$h_L = K_L \frac{V^2}{2g}$$

Thus,

$$K_L = h_L \frac{V^2}{2g}$$

Where,

- h_L = Head Loss(minor loss)
- K_L = Loss Coefficient
- V = Velocity of the fluid
- G = Acceleration due to gravity

5.2.1 Sudden Compression:-

Inlet diameter of the sudden compression region is D = 2.55 cm and the outlet diameter of the sudden compression region is d = 1.54 cm.
In the sudden compression, we consider the outlet diameter (d) for further calculations. Because the outlet diameter(d) is small and the sudden compression depends more on the outlet diameter.

$$A = \frac{\pi d^2}{4}$$

Thus,

$$A = 1.863 cm^2$$

Experimental results : -

For the first flow rate

Reading number	Left tube	Right tube	difference (cm)	difference (m)
1	71.6	25.7	45.9	0.459
2	91.6	68.8	22.8	0.228
3	92.3	68.5	23.8	0.238
		Avg Height difference	30.833	0.30833

From the above table, we can determine the loss coefficient by using velocities from the volumetric flow rate.

Reading number	flow rate (Q) (cm^3/s)	Velocity (m/s)	difference(m)	(K_L)
1	396.301	2.12676	0.459	1.991
2	446.428	2.39577	0.228	0.779
3	417.827	2.24228	0.238	0.929
Average	420.185	2.25494	0.308	1.188

Here K_L is the loss coefficient
To find equivalent length (L_{eq}) of the pipe,

Velocity (m/s)	Reynolds number	Friction factor	Loss Coefficient (KL)	L_eq (m)
2.12676	3.24E+04	2.31E-02	1.991	1.33E+00
2.39577	3.65E+04	2.24E-02	0.779	5.36E-01
2.24228	3.42E+04	2.28E-02	0.929	6.27E-01
2.25494	3.44E+04	2.27E-02	1.190	8.07E-01

For the second flow rate :-

Reading number	Left tube	Right tube	difference (cm)	difference (m)
1	82.2	72.6	9.6	0.096
2	73.7	60	13.7	0.137
3	69.5	56.3	13.2	0.132
4		**Avg height difference**	12.167	0.12167

From the above table, we can determine the loss coefficient by using velocities from the volumetric flow rate.

Reading number	**flow rate (Q) (cm^3/s)**	**Velocity (m/s)**	**Height difference (m)**	K_L
1	236.406	1.26868	0.019	0.232
2	216.763	1.16326	0.020	0.290
3	210.674	1.13059	0.019	0.292
Average	221.281	1.18751	0.01933	0.269

Here K_L is the loss coefficient.
To find equivalent length (L_{eq}) of the pipe.

Velocity (m/s)	**Reynolds number**	**Friction factor**	K_L	L_{eq}**(m)**
1.26868	1.93E+04	2.61E-02	0.232	1.37E-01
1.16326	1.77E+04	2.67E-02	0.290	1.67E-01
1.13059	1.72E+04	2.68E-02	0.292	1.68E-01
1.18751	1.81E+04	2.65E-02	0.269	1.56E-01

Theretical Results - To find the loss coefficient (K_{Lt}) theoretically, we used the formula from the textbook,

$$K_{Lt} = 0.42(1 - \frac{d^2}{D^2})^2$$
$$K_{Lt} = 0.267$$

5.2.2 Sudden Compression:-

Outlet diameter of the sudden expansion region is D = 2.55 cm and the inlet diameter of the sudden expansion region is d = 1.54 cm. In the sudden expansion, we consider the inlet diameter (d) for further calculations. Because the inlet diameter(d) is small and the sudden expansion depends more on the inlet diameter.

$$A = \frac{\pi d^2}{4}$$

Thus,

$$A = 1.863 cm^2$$

For the first flow rate

Reading number	**Left tube**	**Right tube**	**difference (cm)**	**difference (m)**
1	75.7	79.5	3.8	0.038
2	76.8	84.4	7.6	0.076
3	62.1	66.2	4.1	0.041
4		**Avg height difference**	5.167	0.05167

From the above table, we can determine the loss coefficient by using velocities from the volumetric flow rate.

Reading number	**flow rate (Q) (cm^3/s)**	**Velocity (m/s)**	**difference (m)**	K_L
1	396.301	2.12676	0.038	0.165
2	446.428	2.39577	0.076	0.260
3	417.827	2.24228	0.041	0.160
Average	420.185	2.25494	0.05167	0.199

Here K_L is the loss coefficient.
To find equivalent length (L_{eq}) of the pipe.

Velocity (m/s)	**Reynolds number**	**Friction factor**	K_L	L_{eq}**(m)**
2.12676	3.24E+04	2.31E-02	0.165	1.10E-01
2.39577	3.65E+04	2.24E-02	0.260	1.79E-01
2.24228	3.42E+04	2.28E-02	0.160	1.08E-01
2.25494	3.44E+04	2.27E-02	0.199	1.35E-01

For the second flow rate :-

Reading number	**Left tube**	**Right tube**	**difference (cm)**	**difference (m)**
1	73.8	75.7	1.9	0.019
2	89.3	91.3	2.0	0.020
3	80.3	82.2	1.9	0.019
		Avg height difference	1.933	0.01933

From the above table, we can determine the loss coefficient by using velocities from the volumetric flow rate.

Reading number	**flow rate (Q) (cm^3/s)**	**Velocity (m/s)**	**difference (m)**	K_L
1	236.406	1.26868	0.019	0.232
2	216.763	1.16326	0.020	0.290
3	210.674	1.13059	0.019	0.292
Average	221.281	1.18751	0.01933	0.269

Here K_L is the loss coefficient.
To find equivalent length (L_{eq}) of the pipe.

Velocity (m/s)	**Reynolds number**	**Friction factor**	K_L	L_{eq}
1.26868	1.93E+04	2.61E-02	0.232	1.37E-01
1.16326	1.77E+04	2.67E-02	0.290	1.67E-01
1.13059	1.72E+04	2.68E-02	0.292	1.68E-01
1.18751	1.81E+04	2.65E-02	0.269	1.56E-01

Theretical Results - To find the loss coefficient (K_{Lt}) theoretically, we used the formula from the textbook,

$$K_{Lt} = (1 - \frac{d^2}{D^2})^2$$
$$K_{Lt} = 0.404$$

5.2.3 Long Bend

In the bend, head loss occurs due to the change in the direction of the fluid (In this case, fluid is water).
Given that the diameter of the bend is (d) = 1.54 cm.

$$A = \frac{\pi d^2}{4}$$

Thus,

$$A = 1.863cm^2$$

For the first flow rate:

Reading number	Left tube	Right tube	difference (cm)	difference (m)
1	74.5	58.8	15.7	0.157
2	85.0	69.7	15.3	0.153
3	59.3	44.9	14.4	0.144
4		**Avg height difference**	15.133	0.15133

From the above table, we can determine the loss coefficient by using velocities from the volumetric flow rate.

Reading number	flow rate (Q) (cm^3/s)	Velocity (m/s)	difference (m)	K_L
1	396.301	2.12676	0.157	0.681
2	446.428	2.39577	0.153	0.523
3	417.827	2.24228	0.144	0.562
Average	420.185	2.25494	0.15133	0.584

To find equivalent length (L_{eq}) of the pipe.

Velocity (m/s)	Reynolds number	Friction factor	K_L	L_{eq}
2.12676	3.24E+04	2.31E-02	0.681	4.54E-01
2.39577	3.65E+04	2.24E-02	0.523	3.60E-01
2.24228	3.42E+04	2.28E-02	0.562	3.80E-01
2.25494	3.44E+04	2.27E-02	0.584	3.96E-01

For the second flow rate:

Reading number	Left tube	Right tube	difference (cm)	difference (m)
1	70	74	4	0.04
2	84	77.7	6.3	0.063
3	84	77.5	6.5	0.065
		Avg height difference	5.6	0.056

From the above table, we can determine the loss coefficient by using velocities from the volumetric flow rate.

Reading number	flow rate (Q) (cm^3/s)	Velocity (m/s)	difference (m)	K_L
1	236.406	1.26868	0.04	0.488
2	216.763	1.16326	0.063	0.913
3	210.674	1.13059	0.065	0.998
Average	221.281	1.18751	0.056	0.779

To find equivalent length (L_{eq}) of the pipe.

Velocity (m/s)	Reynolds number	Friction factor	K_L	L_{eq}
1	236.406	1.26868	0.04	0.488
2	216.763	1.16326	0.063	0.913
3	210.674	1.13059	0.065	0.998
Average	221.281	1.18751	0.056	0.779

Theoretical

We took the value of the loss coefficient (K_{Lt}) from the online resources,

$$K_{Lt} = 0.3$$

5.2.4 Elbow

In the elbow, head loss occurs due to the change in the direction of the fluid (In this case, fluid is water). The magnitude of the velocity does not change.
Given that the diameter of the elbow is (d) = 1.54 cm.

$$A = \frac{\pi d^2}{4}$$

Thus,

$$A = 1.863cm^2$$

For the first flow rate

Reading number	Left tube	Right tube	difference (cm)	difference (m)
1	75.3	54.8	20.5	0.205
2	73.7	51.6	22.1	0.221
3	87.5	65.3	22.2	0.222
		Avg height difference	21.6	0.216

From the above table, we can determine the loss coefficient by using velocities from the volumetric flow rate.

Reading number	**flow rate (Q) (cm^3/s)**	**Velocity (m/s)**	**difference (m)**	K_L
1	396.301	2.12676	0.205	0.889
2	446.428	2.39577	0.221	0.755
3	417.827	2.24228	0.222	0.866
Average	420.185	2.25494	0.216	0.833

To find equivalent length (L_{eq}) of the pipe.

Velocity (m/s)	**Reynolds number**	**Friction factor**	K_L	L_{eq}
2.12676	3.24E+04	2.31E-02	0.889	5.93E-01
2.39577	3.65E+04	2.24E-02	0.755	5.19E-01
2.24228	3.42E+04	2.28E-02	0.866	5.85E-01
2.25494	3.44E+04	2.27E-02	0.833	5.65E-01

For the second flow rate

Reading number	**Left tube**	**Right tube**	**difference (cm)**	**difference (m)**
1	88.3	79.0	9.3	0.093
2	84.0	74.8	9.2	0.092
3	85.3	77.1	8.2	0.082
		Avg height difference	8.9	0.089

From the above table, we can determine the loss coefficient by using velocities from the volumetric flow rate.

Reading number	**flow rate (Q) (cm^3/s)**	**Velocity (m/s)**	**difference (m)**	K_L
1	236.406	1.26868	0.093	1.134
2	216.763	1.16326	0.092	1.334
3	210.674	1.13059	0.082	1.259
Average	221.281	1.18751	0.089	1.238

To find equivalent length (L_{eq}) of the pipe.

Velocity (m/s)	**Reynolds number**	**Friction factor**	K_L	L_{eq}
1.26868	1.93E+04	2.61E-02	1.134	6.69E-01
1.16326	1.77E+04	2.67E-02	1.334	7.69E-01
1.13059	1.72E+04	2.68E-02	1.259	7.23E-01
1.18751	1.81E+04	2.65E-02	1.238	7.20E-01

Theoretical results

We took the value of the loss coefficient (K_{Lt}) from the online resources,

$$K_{Lt} = 1.1$$

5.2.5 Gate Valve

Given that the diameter of the elbow is (d) = 2.134 cm. Area of the;

$$A = \frac{\pi d^2}{4}$$

Thus,

$$A = 3.577 cm^2$$

For the first flow rate

Reading number	Left tube	Right tube	difference (cm)	difference (m)
1	86.0	21.6	64.4	0.644
2	76.9	20.7	56.2	0.562
3	85.4	26.1	59.3	0.593
		Avg height difference	59.967	0.59967

From the above table, we can determine the loss coefficient by using velocities from the volumetric flow rate

Reading number	flow rate (Q) (cm^3/s)	Velocity (m/s)	difference (m)	K_L
1	396.301	1.10757	0.644	10.300
2	446.428	1.24767	0.562	7.083
3	417.827	1.16773	0.593	8.532
Average	420.185	1.17432	0.59967	8.532

To find equivalent length (L_{eq}) of the pipe.

Velocity (m/s)	Reynolds number	Friction factor	K_L	L_{eq}
1.10757	2.34E+04	2.50E-02	10.300	8.79E+00
1.24767	2.64E+04	2.40E-02	7.083	6.30E+00
1.16773	2.47E+04	2.45E-02	8.532	7.43E+00
1.17432	2.48E+04	2.45E-02	8.532	7.43E+00

For the second flow rate

Reading number	Left tube	Right tube	difference (cm)	difference (m)
1	67.4	42.0	25.4	0.254
2	86.8	62.4	24.4	0.244
3	73.2	48.1	25.1	0.251
		Avg height difference	24.967	0.24967

From the above table, we can determine the loss coefficient by using velocities from the volumetric flow rate

Reading number	flow rate (Q) (cm^3/s)	Velocity (m/s)	difference (m)	K_L
1	236.406	0.660703	0.254	11.416
2	216.763	0.60580	0.244	13.044
3	210.674	0.58879	0.251	14.205
Average	221.281	0.61843	0.24967	12.808

To find equivalent length (L_{eq}) of the pipe.

Velocity (m/s)	Reynolds number	Friction factor	K_L	L_{eq}
0.660703	1.40E+04	2.83E-02	11.416	8.61E+00
0.60580	1.28E+04	2.89E-02	13.044	9.63E+00
0.58879	1.24E+04	2.92E-02	14.205	1.04E+01
0.61843	1.31E+04	2.88E-02	12.808	9.49E+00

Theoretical results

We took the value of the loss coefficient (K_{Lt}) from the online resources,

$$K_{Lt} = 0.2$$

5.3 Error Analysis

Error in Friction Factor

In this section, we calculate the error in the experimental values of the friction factor for all three pipes under both flow rates. The error presented here corresponds to the error in the experimental values.

The formula for absolute error is given by:

$$\text{Absolute Error} = |\text{Measured Value} - \text{Average Value}|$$

Average Absolute Error Calculation:-

We calculate the average absolute error using the formula:

$$\text{Average Absolute Error} = \frac{\epsilon_1 + \epsilon_2 + \epsilon_3}{3}$$

Where:

- ϵ_1, ϵ_2, and ϵ_3 are the absolute errors calculated for each reading.
- The divisor (3) represents the number of readings taken.

5.3.1 Major Losses

Smooth Stainless Steel Pipe

For the First Flow Rate

Reading Number	Friction Factor (f)	Absolute Error
1	0.0387	0.0085
2	0.0262	0.0040
3	0.0269	0.0033
	Average Friction Factor = 0.0302	**Average Absolute Error** = 0.0053

Friction Factor $= 0.0302 \pm 0.0053$
For the Second Flow Rate

Reading Number	Friction Factor (f)	Absolute Error
1	0.049	0.003
2	0.070	0.017
3	0.039	0.053
	Average Friction Factor = 0.053	**Average Absolute Error** = 0.011

Friction Factor $= 0.0532 \pm 0.011$

Roughed Stainless Steel Pipe

For the First Flow Rate

Reading Number	Friction Factor (f)	Absolute Error
1	0.0205	0.002
2	0.0158	0.003
3	0.0186	0.000
	Average Friction Factor = 0.0182	**Average Absolute Error** = 0.001

Friction Factor $= 0.0182 \pm 0.001$
For the Second Flow Rate

Reading Number	Friction Factor (f)	Absolute Error
1	0.008	0.007
2	0.011	0.004
3	0.028	0.013
	Average Friction Factor = 0.015	**Average Absolute Error** = 0.008

Friction Factor $= 0.015 \pm 0.008$

PVC Pipe

For the First Flow Rate

Reading Number	Friction Factor (f)	Absolute Error
1	0.024	0.002
2	0.019	0.003
3	0.022	0.000
	Average Friction Factor = 0.022	**Average Absolute Error** = 0.001

Friction Factor = 0.022 ± 0.001
For the Second Flow Rate

Reading Number	Friction Factor (f)	Absolute Error
1	0.025	0.004
2	0.030	0.001
3	0.032	0.003
	Average Friction Factor = 0.029	**Average Absolute Error** = 0.002

Friction Factor = 0.029 ± 0.002

5.3.2 Minor Losses

Sudden Compression

For the First Flow Rate

Reading Number	Loss coefficient	Absolute Error
1	1.991	0.758
2	0.779	0.454
3	0.929	0.294
	Average Loss Coefficient = 1.233	**Average Absolute Error** = 0.502

Loss Coefficient = 1.233 ± 0.502
For the Second Flow Rate

Reading Number	Loss coefficient	Absolute Error
1	1.170	0.557
2	1.986	0.259
3	2.026	0.299
	Average Loss Coefficient = 1.727	**Average Absolute Error** = 0.372

Loss Coefficient = 1.727 ± 0.372

Sudden Expansion

For the First Flow Rate

Reading Number	Loss coefficient	Absolute Error
1	0.165	0.030
2	0.260	0.065
3	0.160	0.035
	Average Loss Coefficient = 0.195	**Average Absolute Error** = 0.043

Loss Coefficient $= 0.195 \pm 0.043$

For the Second Flow Rate

Reading Number	Loss coefficient	Absolute Error
1	0.232	0.039
2	0.290	0.019
3	0.292	0.021
	Average Loss Coefficient = 0.271	**Average Absolute Error** = 0.0395

Loss Coefficient $= 0.271 \pm 0.0395$

Bend

For the First Flow Rate

Reading Number	Loss coefficient	Absolute Error
1	0.681	0.092
2	0.523	0.066
3	0.562	0.027
	Average Loss Coefficient = 0.589	**Average Absolute Error** = 0.062

Loss Coefficient $= 0.589 \pm 0.062$

For the Second Flow Rate

Reading Number	Loss coefficient	Absolute Error
1	0.487	0.312
2	0.913	0.114
3	0.998	0.199
	Average Loss Coefficient = 0.799	**Average Absolute Error** = 0.208

Loss Coefficient $= 0.799 \pm 0.208$

Elbow

For the First Flow Rate

Reading Number	Loss coefficient	Absolute Error
1	0.889	0.052
2	0.755	0.082
3	0.866	0.029
	Average Loss Coefficient = 0.837	**Average Absolute Error** = 0.054

Loss Coefficient = 0.837 ± 0.054

For the Second Flow Rate

Reading Number	Loss coefficient	Absolute Error
1	1.134	0.108
2	1.334	0.092
3	1.258	0.016
	Average Loss Coefficient = 1.242	**Average Absolute Error** = 0.072

Loss Coefficient = 1.242 ± 0.072

Gate Valve

For the First Flow Rate

Reading Number	Loss coefficient	Absolute Error
1	10.300	1.662
2	7.083	1.555
3	8.532	0.106
	Average Friction Factor = 8.638	**Average Absolute Error** = 1.107

Loss Coefficient = 8.638 ± 1.107

For the Second Flow Rate

Reading Number	Loss coefficient	Absolute Error
1	11.416	1.472
2	13.044	0.156
3	14.205	1.317
	Average Friction Factor = 12.888	**Average Absolute Error** = 0.982

Loss Coefficient = 12.888 ± 0.982

5.4 Percentage Errors

5.4.1 For Major Losses

We'll find the percentage error between the experimental friction factor and the theoretical friction factor using this formula:

$$Error(\%) = (\frac{|f_{exp} - f_{theo}|}{f_{theo}}) \times 100\%$$

Where:

- f_{exp} = experimental friction factor
- f_{theo} = theoretical friction factor.

Flow Rate	Property	Smooth Stainless Steel Pipe	PVC Pipe
Flow Rate 1	$f_{\text{exp,avg}}$	0.0302	0.0220
	$f_{\text{theo,avg}}$	0.0260	0.0231
	$\|f_{\text{exp,avg}} - f_{\text{theo,avg}}\|$	0.0042	0.0011
	Percentage Error	16.153 %	4.762 %
Flow Rate 2	$f_{\text{exp,avg}}$	0.0532	0.0291
	$f_{\text{theo,avg}}$	0.0306	0.0267
	$\|f_{\text{exp,avg}} - f_{\text{theo,avg}}\|$	0.0226	0.0023
	Percentage Error	73.856 %	8.989 %

Table 9: Experimental and Theoretical Friction Factor Comparison

5.4.2 For Minor Losses

We'll find the percentage error between the experimental loss coefficient and the theoretical loss coefficient using this formula:

$$Error(\%) = \frac{|KL_{exp} - KL_{theo}|}{KL_{theo}} \times 100\%$$

Where:

- KL_{exp} = experimental loss coefficient
- KL_{theo} = theoretical loss coefficient

Flow Rate	Property	SC	SE	Bend	Elbow	Gate
Flow Rate 1	KL_{exp}	1.233	0.195	0.589	0.837	8.638
	KL_{theo}	0.267	0.404	0.3	1.1	0.2
	$\|KL_{\text{exp}} - KL_{\text{theo}}\|$	0.966	0.209	0.289	0.263	8.438
	Error(%)	361.798%	51.733%	96.33%	23.909%	4219%
Flow Rate 2	KL_{exp}	1.727	0.271	0.799	1.242	12.888
	KL_{theo}	0.267	0.404	0.3	1.1	0.2
	$\|KL_{\text{exp}} - KL_{\text{theo}}\|$	1.46	0.133	0.499	0.142	12.688
	Error(%)	546.816%	32.921%	166.33%	12.909%	6344%

Table 10: Experimental and Theoretical KL Comparison for Minor Loss Parts

5.5 Sources of Error

1. **Wrong placements of measurement tubes in case of sudden compression**
   In case of sudden compression, the measurement points where the manometer tubes were to be attached were placed at the wrong locations. This was the experimental defect, which led to a vast difference between the experimental readings and the theoretical values.

2. **Air in manometer tubes**
   The formula used for measuring the height difference was

   $$\Delta h = \frac{\Delta P}{\rho g} = h$$

   while the actual formula is

   $$\Delta h = \frac{\Delta P}{\rho g} = h(1 - \frac{P_a}{P_w})$$

   which accounts for the pressure due to air in the manometer tube as well. The second term in the second formula is ignored because it is negligible in comparison to the other, but neglecting the other term may lead to deviations from the theoretical values.

3. **Response error -**
   When we measured the time for the specific height reached in the tank to determine the flow rate, the time was measured manually with the help of a stopwatch. So, it can lead to a delay in time measurement.

4. **Instrumental inaccuracies -**
   There were minor defects in the experimental apparatus, such as pipes being bent and valves being loose, which could eventually cause uncertainties in the results.

5. **Parallax error -**
   This error is caused when the observer takes measurements from an angle, resulting in differences from the actual reading.

6. **Imperfections in fluid -**
The fluid used to experiment was water. However, some impurities, such as dirt and air bubbles in the fluid, could contribute to the total error.

7. **Least count errors -**
The number called least count is associated with every measuring instrument; errors arising from the least count are termed as least count errors. The contribution of the least count errors to the total error depends upon the instrument's resolution.

6 Observation

Minor losses occur due to changes in the direction of a fluid. In the minor losses, there is less error in the elbow and a significant error in the gate valve. The sudden compression also has a high error because the manometer is attached to the wrong place. The manometer should be connected to the input and output pipes to find the correct minor loss (or head loss). The experimental loss coefficient values differ from the theoretical values due to the turbulent flow or the flow disturbance.

7 Conclusion

The results of this experiment show that the major losses in the straight tube can slightly depend on the flow rate of the fluid and the type of material in the tube. Rougher material exhibits higher frictional loss. Minor losses were found to vary depending on the system configuration and geometry and the flow rate. They were also associated with components like bends, valves, expansions, and contractions. Both experimental and theoretical results are different from each other. This error occurs due to some reasons like instrumental error, human error, etc.

8 Acknowlegements

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